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# **Software Requirements Specification**

**for**

## ***Sandbox AI-Powered Online Assessment and Intelligent Proctoring Platform***

**Version 1.0**

**Prepared by Project Team 5**

|                                       |                     |
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## Revision History

| Name   | Date        | Reason For Changes   | Version |
|--------|-------------|----------------------|---------|
| Team 5 | 21-Jul-2026 | Initial SRS Creation | 1.0     |
|        |             |                      |         |

# **1. Introduction**

## **1.1 Purpose**

The purpose of this document is to provide a comprehensive description of Sandbox, an AI-Powered Online Assessment and Intelligent Proctoring Platform. This document specifies the functional and non-functional requirements of the system, including assessment management, candidate management, browser-based proctoring, subscription management, and report generation. It serves as a reference for the design, development, testing, and maintenance of the system.

## **1.2 Intended Audience and Reading Suggestions**

This document is intended for developers, testers, project managers, organizations, and system administrators.

For developers, this document provides the functional and technical requirements required to implement the system. Testers can use it to prepare test cases and validate the application against the specified requirements. Organizations can understand the capabilities of the platform and its assessment workflow, while project managers can use it to monitor project scope and progress.

Readers are advised to begin with the Introduction and Overall Description before referring to the detailed system features and non-functional requirements.

## **1.3 Product Scope**

Sandbox is a cloud-based Software-as-a-Service (SaaS) platform that enables organizations to conduct secure online assessments. The platform supports MCQ, coding, and mixed examinations with browser-based proctoring, candidate management, automated evaluation, subscription management, payment integration, report generation, and certificate generation.

The system follows a multi-tenant architecture, ensuring that each organization's assessments, candidates, reports, and question banks remain isolated and accessible only to authorized users.

## **1.4 References**

- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
- Spring Boot Documentation — <https://docs.spring.io/spring-boot/>
- React Documentation — <https://react.dev/>
- MySQL Documentation — <https://dev.mysql.com/doc/>
- MongoDB Documentation — <https://www.mongodb.com/docs/>

- Razorpay Developer Documentation — <https://razorpay.com/docs/>
- RFC 7519 – JSON Web Token (JWT)

## 2. Overall Description

### 2.1 Product Perspective

Sandbox is a cloud-based Software-as-a-Service (SaaS) platform developed to simplify and secure the process of conducting online assessments. The platform enables organizations to create, manage, and evaluate MCQ, coding, and mixed examinations through a centralized web application. It follows a multi-tenant architecture, ensuring complete isolation of data between organizations while allowing the Super Admin to manage the platform centrally. The system integrates secure authentication, browser-based proctoring, payment management, and reporting to provide an end-to-end assessment solution.

### 2.2 Product Functions

Sandbox provides the following major functionalities:

- User registration, email verification, and secure authentication.
- Organization registration and subscription management.
- Creation and management of question banks.
- Support for MCQ, coding, and mixed assessments.
- Candidate registration and assessment assignment.
- Browser-based proctoring with activity monitoring.
- Automatic evaluation and result generation.
- Dashboard and analytical reports for organizations.
- Certificate generation based on organization settings.
- Secure payment processing through Razorpay.

### 2.3 User Classes and Characteristics

The system consists of three primary user classes:

**Super Admin:** Responsible for managing organizations, subscriptions, payments, and monitoring the overall platform.

**HR (Organization Admin):** Responsible for managing question banks, assessments, candidates, reports, certificates, and organization-specific settings. HR users can access only the data belonging to their own organization.

**Candidate:** Registers on the platform, verifies their email address, attempts assigned assessments, and views results or certificates based on the permissions provided by the organization.

## 2.4 Operating Environment

Sandbox is a web-based application that supports modern browsers including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari. The frontend is developed using React, while the backend is built with Spring Boot. The system uses MySQL for relational data, MongoDB for proctoring logs, and RabbitMQ for asynchronous communication. It can be deployed using Docker containers or cloud platforms such as AWS.

## 2.5 Design and Implementation Constraints

The system is designed using a multi-tenant architecture to ensure complete data isolation between organizations. JWT-based authentication, email verification, and BCrypt password encryption are used to secure user accounts. Browser-based proctoring requires permissions for webcam, microphone, and fullscreen access. Subscription management and payment processing are implemented using Razorpay. The platform must support multiple programming languages for coding assessments while maintaining secure and reliable execution.

## 2.6 User Documentation

The following documentation will be provided with the system:

- User Guide for Candidates
- HR Administration Guide
- Super Admin Guide
- Installation and Deployment Guide
- API Documentation

## 2.7 Assumptions and Dependencies

The system assumes that users have a stable internet connection and use a supported web browser with webcam and microphone access. Organizations are responsible for managing their own candidates, assessments, and question banks. The availability of email services and the payment gateway is required for account verification and subscription management. Code execution and AI-based evaluation services will depend on the technologies finalized during implementation.

# 3. External Interface Requirements

## 3.1 User Interfaces

*Sandbox provides a responsive and user-friendly web interface designed for Super Admins, HRs, and Candidates. The interface follows a consistent layout with intuitive navigation, dashboards, forms, and validation messages. Users interact with the system through web pages for registration, login, assessment management, candidate management, report generation, and examination. During assessments, candidates are provided with a distraction-free interface featuring a countdown timer, navigation panel, coding editor (for coding questions), and real-time proctoring indicators.*

## 3.2 Hardware Interfaces

*The system is accessible through desktop and laptop devices equipped with a modern web browser. For browser-based proctoring, candidates must have a functional webcam, microphone, and stable internet connection. The system also requires access to the device's display to monitor fullscreen mode, screen activity, and multiple monitor usage during assessments.*

## 3.3 Software Interfaces

*Sandbox integrates with several software components to provide its functionality. The frontend is developed using **React**, while the backend is implemented using **Spring Boot**. **MySQL** is used for relational data storage, **MongoDB** stores proctoring-related logs, and **RabbitMQ** enables asynchronous communication between services. The platform also integrates with **Razorpay** for payment processing and uses SMTP-based email services for account verification and notifications.*

## 3.4 Communications Interfaces

*The system communicates over secure **HTTPS** connections to ensure safe data transmission. RESTful APIs are used for communication between the frontend and backend services. Email notifications are delivered using SMTP services for account verification, password reset, and assessment-related updates. Internal service communication is handled through RabbitMQ message queues where applicable.*

# 4. System Features

*This section describes the major functional features of the Sandbox platform. It outlines the core modules of the system along with their purpose, priority, user interactions, and functional requirements. These features define the expected behavior of the system and serve as the basis for design, implementation, and testing. Each module is presented independently to provide a clear understanding of its functionality and its contribution to the overall system.*

## 4.1 User Authentication

### 4.1.1 Description and Priority

#### Description

The User Authentication module provides secure access to the Sandbox platform for Super Admins, HRs, and Candidates. It supports user registration, email verification, login, password recovery, logout, and role-based access control. The module ensures that only authenticated users can access the system and perform actions according to their assigned roles.

### 4.1.2 Stimulus/Response Sequences

| Stimulus                            | System Response                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------|
| User submits registration details   | The system validates the information, creates the account, and sends an email verification link. |
| User clicks the verification link   | The system verifies the email address and activates the account.                                 |
| User enters valid login credentials | The system authenticates the user and redirects them to the appropriate dashboard.               |
| User enters invalid credentials     | The system denies access and displays an appropriate error message.                              |
| User requests password reset        | The system sends an OTP or password reset link to the registered email address.                  |
| User logs out                       | The system terminates the current session and redirects the user to the login page.              |

### 4.1.3 Functional Requirements

- The system shall allow users to register using a valid email address.
- The system shall send an email verification link after successful registration.
- The system shall activate user accounts only after successful email verification.
- The system shall allow registered users to log in using valid credentials.
- The system shall prevent access to unauthorized users.
- The system shall allow users to reset their password through the registered email address.
- The system shall maintain separate access privileges for Super Admins, HRs, and Candidates.
- The system shall allow authenticated users to securely log out of the application.

## 4.2 Organization Management Module

### 4.2.1 Description and Priority

The Organization Management module enables the Super Admin to register and manage organizations on the Sandbox platform. Each organization is assigned a dedicated HR account responsible for managing assessments, candidates, question banks, reports, and organization-specific settings. The module ensures complete data isolation, allowing organizations to access only their own resources while preventing unauthorized access to data belonging to other organizations.

### 4.2.2 Stimulus/Response Sequences

#### Stimulus

Super Admin registers a new organization

HR logs into the platform

HR updates organization details

HR attempts to access another organization's data

Super Admin deactivates an organization

#### System Response

The system creates the organization and its dedicated HR account.

The system grants access to organization-specific resources and dashboard.

The system validates and saves the updated information.

The system denies access and displays an authorization error.

The system disables access for all users associated with that organization.

### 4.2.3 Functional Requirements

- The system shall allow the Super Admin to register new organizations.
- The system shall create one HR account for each registered organization.
- The system shall allow HR users to manage only their organization's data.
- The system shall prevent organizations from accessing data belonging to other organizations.
- The system shall allow the Super Admin to activate or deactivate organizations.
- The system shall allow HR users to update organization profile information.
- The system shall maintain separate records for each organization in the database.
- The system shall display organization-specific dashboards after successful login.

## 4.3 Subscription & Payment Management Module

### 4.3.1 Description and Priority

The Subscription & Payment Management module enables organizations to subscribe to suitable plans based on the number of assessments and candidates they require. It allows organizations to view available plans, purchase or renew subscriptions, and monitor their usage. Payment transactions are processed through Razorpay, while the system maintains subscription details and validates plan limits before allowing access to premium features.

### 4.3.2 Stimulus/Response Sequences

#### Stimulus

HR selects a subscription plan

HR proceeds with payment

Payment is successful

Payment fails

#### System Response

The system displays the plan details and pricing.

The system redirects the user to the Razorpay payment gateway.

The system activates or renews the subscription and updates the plan details.

The system displays an appropriate message and keeps the current subscription unchanged.

**Stimulus**

Organization reaches its subscription limit

**System Response**

The system prevents further operations and prompts the HR to upgrade or renew the subscription.

**4.3.3 Functional Requirements**

- The system shall allow organizations to view available subscription plans.
- The system shall allow organizations to subscribe to or renew a plan.
- The system shall process payments through the integrated payment gateway.
- The system shall activate the selected subscription after successful payment.
- The system shall maintain the subscription history of each organization.
- The system shall restrict the number of assessments and candidates according to the active subscription plan.
- The system shall notify organizations before the subscription expires.
- The system shall allow the Super Admin to manage available subscription plans.

**4.4 Question Bank Management Module****4.4.1 Description and Priority**

The Question Bank Management module enables HR users to create, organize, update, and manage questions for their organization's assessments. The system supports MCQ and coding questions, allowing HRs to define options, sample test cases, hidden test cases, difficulty levels, and other question-specific details. Each organization maintains an independent question bank, ensuring that questions remain private and cannot be accessed by other organizations.

**4.4.2 Stimulus/Response Sequences****Stimulus**

HR creates a new question

HR edits an existing question

HR deletes a question

HR searches or filters questions

HR attempts to access another organization's question bank

**System Response**

The system validates the input and stores the question in the organization's question bank.

The system updates the question details and saves the changes.

The system removes or marks the question as inactive based on system settings.

The system displays matching questions based on the selected criteria.

The system denies access and displays an authorization message.

**4.4.3 Functional Requirements**

- The system shall allow HR users to create MCQ and coding questions.

- The system shall allow HR users to add multiple options for MCQ questions.
- The system shall allow HR users to define sample and hidden test cases for coding questions.
- The system shall allow HR users to specify the difficulty level and marks for each question.
- The system shall allow HR users to organize questions based on custom categories or folders.
- The system shall allow HR users to edit or delete existing questions.
- The system shall provide search and filter functionality within the question bank.
- The system shall ensure that each organization's question bank remains isolated from other organizations.
- The system shall prevent unauthorized users from modifying question bank data.
- The system shall allow questions to be reused across multiple assessments within the same organization.

## 4.5 Assessment Management Module

### 4.5.1 Description and Priority

**The Assessment Management module enables HR users to create, configure, schedule, and manage online assessments. The system supports MCQ, coding, and mixed assessments with configurable settings such as duration, passing criteria, negative marking, question randomization, and attempt limits. HR users can assign assessments to eligible candidates and monitor their status throughout the assessment lifecycle.**

### 4.5.2 Stimulus/Response Sequences

| Stimulus                                | System Response                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------|
| HR creates a new assessment             | The system creates the assessment and saves its configuration.                          |
| HR adds questions to the assessment     | The system associates the selected questions with the assessment.                       |
| HR schedules the assessment             | The system stores the date, time, and duration of the assessment.                       |
| HR assigns candidates to the assessment | The system grants access only to the selected candidates.                               |
| HR modifies assessment settings         | The system validates and updates the assessment details.                                |
| HR publishes the assessment             | The system makes the assessment available to eligible candidates at the scheduled time. |

### 4.5.3 Functional Requirements

- The system shall allow HR users to create MCQ, coding, and mixed assessments.
- The system shall allow HR users to define the assessment title, description, and duration.
- The system shall allow HR users to configure the start time and end time of an assessment.
- The system shall allow HR users to add or remove questions before publishing the assessment.
- The system shall allow HR users to set the maximum number of attempts for an assessment.
- The system shall allow HR users to configure passing criteria and negative marking.

- The system shall allow HR users to enable or disable question randomization.
- The system shall allow HR users to assign assessments only to selected candidates.
- The system shall prevent candidates from accessing assessments before the scheduled start time or after the end time.
- The system shall allow HR users to edit or cancel unpublished assessments.

## 4.6 Candidate Management Module

### 4.6.1 Description and Priority

The Candidate Management module enables HR users to manage candidate information and assign assessments to eligible candidates. Candidates can register on the platform, verify their email address, and access only the assessments assigned to them. The module maintains candidate profiles, tracks assessment assignments, and ensures that candidates can attempt assessments only within the permitted schedule.

### 4.6.2 Stimulus/Response Sequences

| Stimulus                                | System Response                                                              |
|-----------------------------------------|------------------------------------------------------------------------------|
| Candidate registers on the platform     | The system creates a candidate account and sends an email verification link. |
| Candidate verifies the email address    | The system activates the candidate account.                                  |
| HR assigns an assessment to a candidate | The system grants the candidate access to the assigned assessment.           |
| Candidate logs into the dashboard       | The system displays only the assessments assigned to the candidate.          |
| HR updates or removes a candidate       | The system saves the changes and updates the candidate records accordingly.  |

### 4.6.3 Functional Requirements

- The system shall allow candidates to register using a valid email address.
- The system shall verify the candidate's email before activating the account.
- The system shall allow HR users to view and manage candidate records within their organization.
- The system shall allow HR users to assign assessments to selected candidates.
- The system shall ensure that candidates can view only the assessments assigned to them.
- The system shall prevent candidates from accessing assessments that have not been assigned.
- The system shall allow HR users to update or remove candidate records.
- The system shall maintain the assessment assignment history of each candidate.
- The system shall notify candidates when a new assessment is assigned.
- The system shall ensure that candidate information remains isolated within the respective organization.

## 4.7 Online Examination Module

### 4.7.1 Description and Priority

The Online Examination module enables candidates to securely attempt assigned assessments through a web-based interface. It provides an interactive environment for answering MCQ and coding questions, manages assessment timing, records responses, executes coding solutions against predefined test cases, and automatically submits the assessment upon completion or when the allotted time expires.

### 4.7.2 Stimulus/Response Sequences

| Stimulus                                                         | System Response                                                      |
|------------------------------------------------------------------|----------------------------------------------------------------------|
| Candidate starts an assigned assessment                          | The system verifies eligibility and opens the examination interface. |
| Candidate answers MCQ or coding questions                        | The system saves the responses automatically during the assessment.  |
| Candidate submits the assessment                                 | The system validates the submission and stores the responses.        |
| Assessment time expires                                          | The system automatically submits the assessment.                     |
| Candidate attempts to access an unassigned or expired assessment | The system denies access and displays an appropriate message.        |

### 4.7.3 Functional Requirements

- The system shall allow candidates to access only their assigned assessments.
- The system shall provide a secure interface for attempting MCQ, coding, and mixed assessments.
- The system shall display the remaining assessment time throughout the examination.
- The system shall automatically save candidate responses during the assessment.
- The system shall allow candidates to navigate between questions before submission.
- The system shall compile and execute coding solutions against predefined sample and hidden test cases.
- The system shall automatically submit the assessment when the allotted time expires.
- The system shall prevent candidates from attempting the same assessment more than the permitted number of times.
- The system shall store all submitted responses securely for evaluation.
- The system shall display an acknowledgment message after successful submission.

## 4.8 Browser-Based Proctoring Module

### 4.8.1 Description and Priority

The Browser-Based Proctoring module continuously monitors candidates during online assessments to help maintain examination integrity. It utilizes browser-based technologies to detect suspicious activities such as multiple face detection, mobile phone usage, tab switching, copy-paste attempts, fullscreen exit, browser resizing, excessive background noise, and multiple monitor usage. The system records proctoring events, calculates a risk score based on detected violations, and generates logs that can be reviewed by HR after the assessment.

#### 4.8.2 Stimulus/Response Sequences

| Stimulus                                                 | System Response                                                                             |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Candidate starts the assessment                          | The system requests access to the webcam and microphone and initializes browser monitoring. |
| A suspicious activity is detected                        | The system records the event and updates the candidate's risk score.                        |
| Candidate switches browser tabs or exits fullscreen mode | The system logs the violation and continues monitoring.                                     |
| Mobile phone or multiple faces are detected              | The system records the event and increases the risk score.                                  |
| Assessment is completed                                  | The system stores all proctoring logs and generates a monitoring summary for HR review.     |

#### 4.8.3 Functional Requirements

- The system shall request webcam and microphone permissions before the assessment begins.
- The system shall monitor candidates throughout the assessment using browser-based proctoring mechanisms.
- The system shall detect and record multiple face detection events.
- The system shall detect and record mobile phone usage during the assessment.
- The system shall detect tab switching and browser focus changes.
- The system shall detect copy and paste attempts within the examination window.
- The system shall detect exits from fullscreen mode.
- The system shall detect browser window resizing during the assessment.
- The system shall detect excessive background noise using the microphone.
- The system shall detect the use of multiple monitors whenever supported by the browser.
- The system shall calculate a cumulative risk score based on detected violations.
- The system shall maintain detailed proctoring logs for each assessment attempt.
- The system shall allow HR users to review proctoring events and risk scores after the assessment is completed.
- The system shall securely store proctoring data for future reference and analysis.

### 4.9 Reports & Analytics Module

#### 4.9.1 Description and Priority

The Reports & Analytics module provides HR users with comprehensive insights into assessment performance and candidate results. It enables organizations to view assessment statistics, individual candidate performance, proctoring summaries, and overall assessment outcomes through interactive reports and dashboards. These reports assist HR users in evaluating candidate performance and making informed recruitment decisions.

#### 4.9.2 Stimulus/Response Sequences

| Stimulus                                      | System Response                                                               |
|-----------------------------------------------|-------------------------------------------------------------------------------|
| HR requests an assessment report              | The system generates and displays the assessment report.                      |
| HR views a candidate's performance            | The system displays the candidate's score, responses, and proctoring summary. |
| HR filters reports by assessment or candidate | The system displays the filtered results.                                     |
| HR downloads a report                         | The system generates the report in the selected format for download.          |
| HR accesses the analytics dashboard           | The system displays assessment statistics and performance metrics.            |

#### 4.9.3 Functional Requirements

- The system shall generate assessment reports for completed assessments.
- The system shall display individual candidate scores and performance summaries.
- The system shall provide assessment statistics, including participation and completion details.
- The system shall display proctoring summaries and risk scores for each candidate.
- The system shall allow HR users to search and filter reports.
- The system shall allow HR users to download assessment reports.
- The system shall provide graphical and tabular representations of assessment performance.
- The system shall restrict report access to authorized users within the respective organization.
- The system shall maintain historical assessment reports for future reference.

### 4.10 Certificate Management Module

#### 4.10.1 Description and Priority

The Certificate Management module enables organizations to generate and issue digital certificates for candidates who successfully complete assessments, based on the organization's configuration. HR users can enable or disable certificate generation for individual assessments, while eligible candidates can access and download their certificates after the assessment results are published.

#### 4.10.2 Stimulus/Response Sequences

| <b>Stimulus</b>                                     | <b>System Response</b>                                                |
|-----------------------------------------------------|-----------------------------------------------------------------------|
| HR enables certificate generation for an assessment | The system marks the assessment as certificate-enabled.               |
| Assessment results are published                    | The system identifies eligible candidates for certificate generation. |
| Eligible candidate accesses the certificate section | The system displays the available certificate for download.           |
| Candidate downloads the certificate                 | The system generates and provides the digital certificate.            |
| HR disables certificate generation                  | The system does not generate certificates for that assessment.        |

### 4.10.3 Functional Requirements

- The system shall allow HR users to enable or disable certificate generation for an assessment.
- The system shall generate digital certificates for eligible candidates after assessment completion.
- The system shall allow candidates to view and download their certificates.
- The system shall include relevant assessment and candidate information on the certificate.
- The system shall prevent ineligible candidates from accessing certificates.
- The system shall maintain a record of issued certificates.
- The system shall restrict certificate management to authorized HR users within the respective organization.

## 4.11 Notification Module

### 4.11.1 Description and Priority

The Notification module is responsible for delivering timely updates and alerts to users regarding important system activities. It sends email notifications for account verification, password reset, assessment assignments, assessment schedules, subscription updates, and result announcements. The module ensures that users receive relevant information to effectively interact with the platform.

### 4.11.2 Stimulus/Response Sequences

| <b>Stimulus</b>                         | <b>System Response</b>                                                 |
|-----------------------------------------|------------------------------------------------------------------------|
| User registers on the platform          | The system sends an email verification link or OTP.                    |
| User requests password recovery         | The system sends a password reset link or OTP to the registered email. |
| HR assigns an assessment to a candidate | The system notifies the candidate about the new assessment.            |
| Assessment results are published        | The system notifies the candidate that the results are available.      |
| Subscription is nearing expiration      | The system sends a reminder notification to the organization's HR.     |

### 4.11.3 Functional Requirements

- The system shall send email verification notifications during user registration.
- The system shall send password reset notifications upon user request.
- The system shall notify candidates when an assessment is assigned to them.
- The system shall notify candidates when assessment results are published.
- The system shall notify HR users about subscription renewal or expiration.
- The system shall ensure that notifications are delivered only to the intended recipients.
- The system shall maintain a record of sent notifications for auditing purposes.
- The system shall allow the system administrator to configure notification templates.

## 4.12 Super Admin Module

### 4.12.1 Description and Priority

The Super Admin module provides centralized control over the Sandbox platform. It enables the Super Admin to manage organizations, HR accounts, subscription plans, and overall system operations. The Super Admin has access to all organizations and can monitor platform activities, configure global settings, and ensure the secure and efficient operation of the system.

### 4.12.2 Stimulus/Response Sequences

| Stimulus                                 | System Response                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------|
| Super Admin logs into the platform       | The system displays the Super Admin dashboard with complete administrative controls. |
| Super Admin registers a new organization | The system creates the organization and assigns its HR account.                      |
| Super Admin updates subscription plans   | The system saves the updated subscription details.                                   |
| Super Admin views system statistics      | The system displays platform-wide reports and analytics.                             |
| Super Admin deactivates an organization  | The system disables access for all users associated with that organization.          |

### 4.12.3 Functional Requirements

- The system shall allow the Super Admin to manage organizations registered on the platform.
- The system shall allow the Super Admin to create, update, activate, or deactivate organizations.
- The system shall allow the Super Admin to manage HR accounts for each organization.
- The system shall allow the Super Admin to create and manage subscription plans.
- The system shall allow the Super Admin to monitor platform-wide assessment and user statistics.
- The system shall allow the Super Admin to access reports across all organizations.

- The system shall restrict Super Admin privileges to authorized users only.
- The system shall maintain audit logs for critical administrative actions performed by the Super Admin.

## 5. Other Nonfunctional Requirements

This section describes the non-functional requirements of the Sandbox platform. These requirements define the expected performance, safety, security, and quality characteristics of the system to ensure reliable, secure, and efficient operation.

### 5.1 Performance Requirements

*The Sandbox platform shall provide a responsive and efficient user experience during all system operations. User authentication and dashboard loading should complete within **3 seconds** under normal network conditions. Assessment pages should load within **5 seconds**, and the system shall support multiple concurrent candidates based on the organization's active subscription plan. Candidate responses shall be saved automatically at regular intervals to minimize data loss, and the system shall automatically submit assessments when the allotted time expires.*

### 5.2 Safety Requirements

*The system shall minimize the risk of data loss and unauthorized actions during online assessments. Automatic response saving and timed auto-submission shall help preserve candidate work in case of unexpected interruptions. Access to assessments shall be restricted to authorized candidates within the scheduled examination period. All critical operations, including assessment submissions and administrative actions, shall be logged to support auditing and recovery when required.*

### 5.3 Security Requirements

*The Sandbox platform shall implement appropriate security mechanisms to protect user accounts and organizational data. User authentication shall be secured using **JWT-based authentication**, and passwords shall be stored using **BCrypt hashing**. All communication between clients and the server shall use **HTTPS**. Role-based access control shall ensure that users can access only the resources permitted by their assigned role. Organizations shall operate in isolated environments to prevent unauthorized access to data belonging to other organizations. Proctoring logs and assessment data shall be securely stored and accessible only to authorized users.*

### 5.4 Software Quality Attributes

*The Sandbox platform shall be designed to provide high levels of reliability, usability, maintainability, scalability, and availability. The system shall offer an intuitive user interface for candidates, HR users, and administrators. Its modular architecture shall simplify maintenance and future enhancements. The platform shall support deployment on cloud infrastructure and be capable of scaling to accommodate increasing numbers of organizations and candidates. The application shall remain compatible with modern web browsers, including Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari.*

## 5.5 Business Rules

*The Sandbox platform shall operate according to the following business rules:*

- *Each organization shall have a dedicated HR account responsible for managing its assessments and candidates.*
- *Organizations shall access only their own data, ensuring complete data isolation.*
- *Candidates shall be allowed to attempt only the assessments assigned to them.*
- *An assessment shall be available only during its scheduled time window.*
- *The maximum number of assessment attempts shall be determined by the HR while creating the assessment.*
- *Organizations shall be restricted by the limits defined in their active subscription plan.*
- *Certificates shall be generated only if certificate generation is enabled for the assessment.*

## 6. Other Requirements

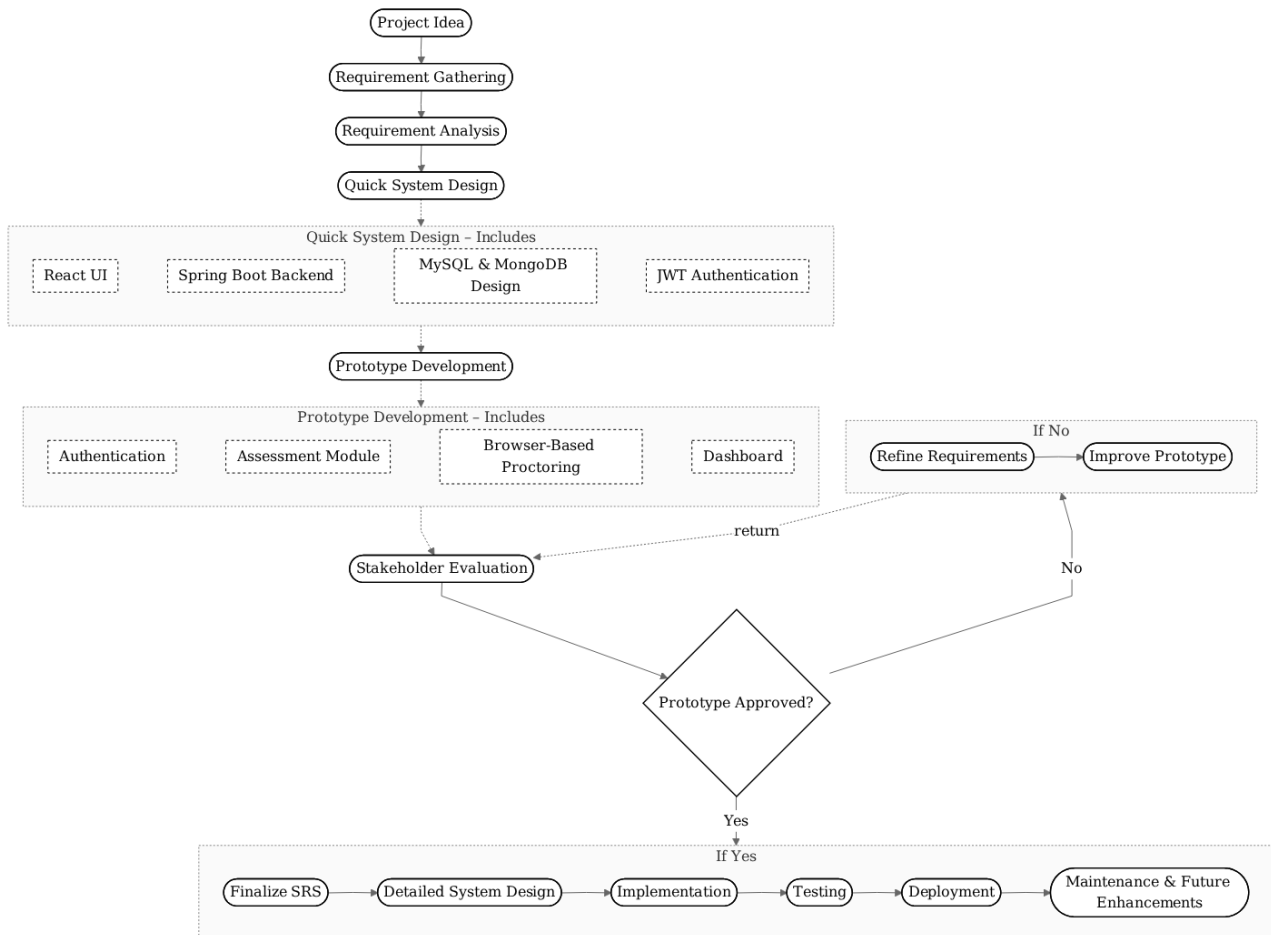
The Sandbox platform shall support future enhancements without requiring significant architectural changes. The system shall follow a modular design to facilitate maintenance and scalability. All system logs, assessment records, and proctoring data shall be stored securely to support auditing and future analysis. The platform shall be deployable on cloud infrastructure and designed to accommodate increasing numbers of organizations and candidates.

## Appendix A: Glossary

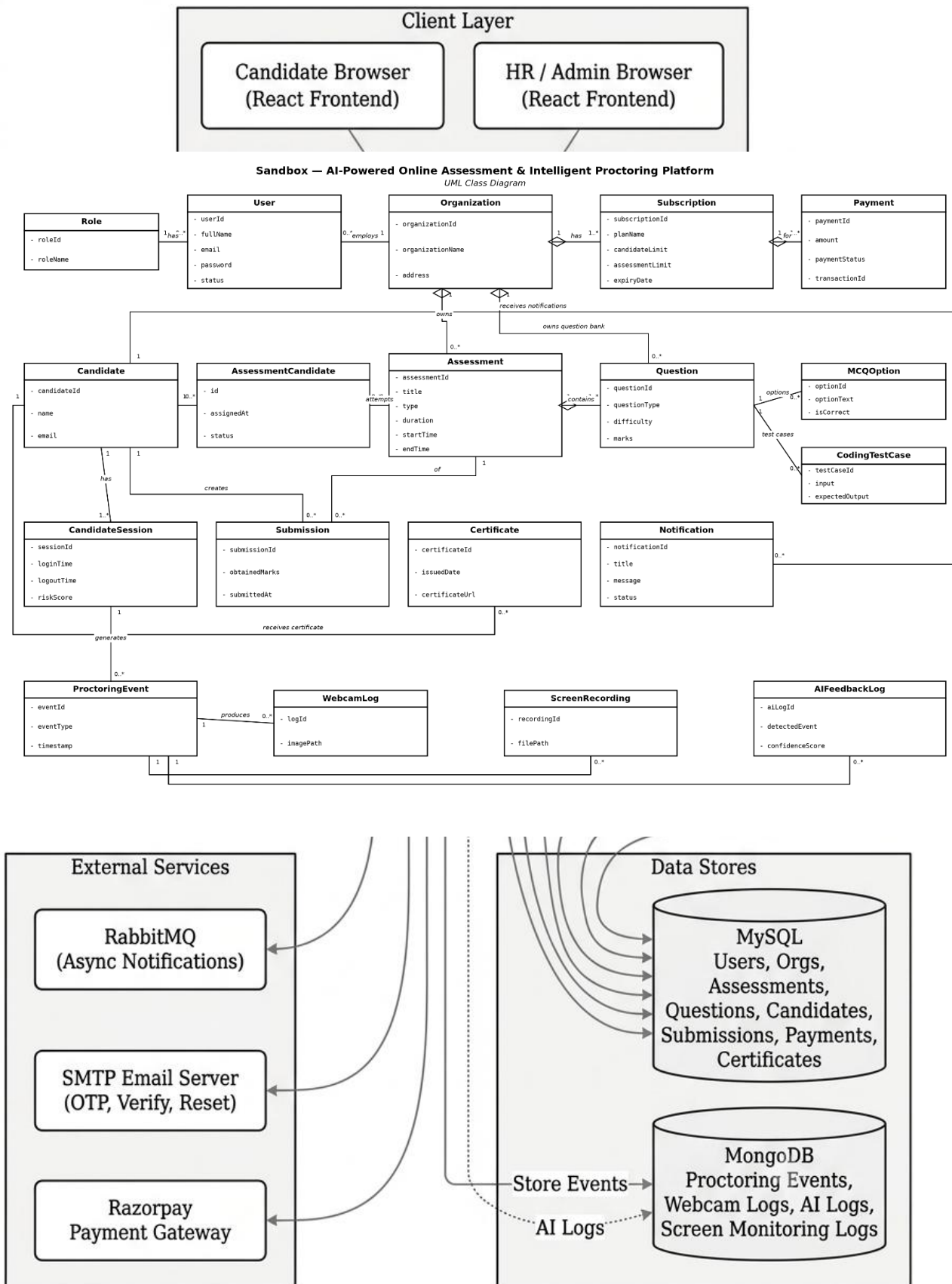
| <b>Term</b> | <b>Description</b>                 |
|-------------|------------------------------------|
| AI          | Artificial Intelligence            |
| HR          | Human Resources                    |
| JWT         | JSON Web Token                     |
| OTP         | One-Time Password                  |
| MCQ         | Multiple Choice Question           |
| SaaS        | Software as a Service              |
| HTTPS       | Hypertext Transfer Protocol Secure |
| API         | Application Programming Interface  |
| UI          | User Interface                     |
| CRUD        | Create, Read, Update, Delete       |
| IDE         | Integrated Development Environment |
| SMTP        | Simple Mail Transfer Protocol      |
| RBAC        | Role-Based Access Control          |
| ORM         | Object Relational Mapping          |

## Appendix B: Analysis Models

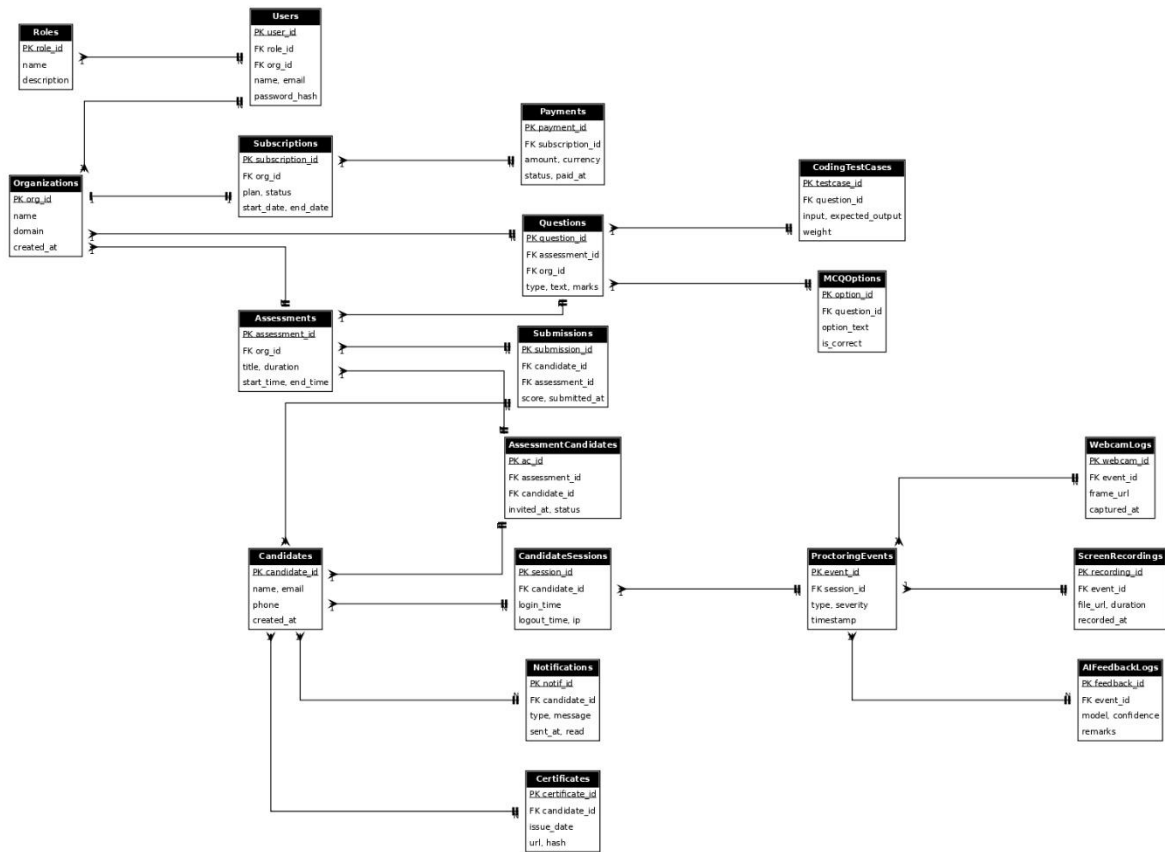
### • Software Development Approach in Our System



## • Class Diagram



- ER Diagram



## **Appendix C: To Be Determined List**

*At the time of preparing this Software Requirements Specification, the following items are subject to future implementation or enhancement:*

***TBD ID Description***

*TBD-01 AI model selection for browser-based proctoring*

*TBD-02 Cloud deployment environment (AWS/Docker)*

*TBD-03 Advanced analytics and reporting enhancements*

*TBD-04 Additional programming languages for coding assessments*

*TBD-05 Future integration with third-party recruitment platforms*